

Course Syllabus

MATH 8440 - Combinatorics

Fall semester, 2026

Course Website. The majority of the course material can be found at samspiro.org, or if this link does not work at <https://samspiro.xyz/Teaching/Classes/GradCom/>. There you can find course notes, homework assignments, and an anonymous feedback form.

Textbook

This course will use my own custom notes at samspiro.org. Other resources that closely follow our notes include the textbook *Extremal Problems for Finite Sets* by Frankl and Tokushige, as well as [course notes](#) by Liu.

Class Time and Location: Tuesday/Thursday 2:15pm to 3:30pm; Aderhold Learning Center Room 230.

Instructor: Dr. Sam Spiro

Email: sspiro@gsu.edu.
Office: 25 Park Place, Room 1319.
Office Hours: Tuesday 1-2 and by appointment.

Course Content

This course will cover various portions of the following two main topics:

- Extremal Set Theory (antichains, intersecting families, sunflowers, VC dimension).
- Discrete Geometry and Topological Methods (convex geometry, the Borsuk-Ulam Theorem, the KKM Theorem, simplicial complexes)

The exact content covered will be based on the interests of the students.

Course Evaluation

Your course grade will be determined as follows:

- **Homework (70%)**
 - There will be weekly homework assignments due each Thursday at 2 starting in week 2.
 - Homeworks should be dropped off in my mailbox on the 13th floor of 25 park place by 2:10 each Thursday.
 - All grading and extension requests for homeworks should be made directly to the TA Shengze Wang at swang85@student.gsu.edu.
 - The homeworks will be posted to the course websites, with “???” indicating that the assignment has not yet been finalized (in case you want to get a head start on what is likely to be the next assignment).
 - I am intending for the homework to be challenging enough that you won’t necessarily be able to do all of it every week, but that you should be able to do most of it most weeks.
 - If you find the homework too challenging, please come see me during office hours (Note: I will not reply to emails about homework problems). If you find the homework too easy, you are encouraged to try some of the unassigned exercises in the notes.
- **Final Exam (30%)** We will have an **in person, closed note** final for this course on December 10th from 1:30 to 4:00.

AI Policy

For homeworks, you may use AI **only** in the following two ways:

- Asking for a hint to a problem after thinking about the problem for at least half an hour. Your request for a hint must include a full writeup of your current thoughts about how to solve the problem.
- Asking it to look over your arguments for corrections while explicitly asking it NOT to correct any mistakes it finds (only to point them out to you).

In both of these cases you must declare in your homework how AI was used for the assignment (no declaration is needed if AI was not used at all).

Technology in Class

You can use whatever devices you'd like during class as long as you do not use them in the front row so that students who might be distracted by your screens can sit in front of you.

Accommodations

Students who wish to request an accommodation may do so by registering with the Access and Accommodation Center (AACE). Students may only be accommodated if the instructor receives an email with the notification of approved accommodations from AACE. Note: Accommodations are not retroactive and will be applicable following the date in which they are received.

Academic Integrity Policy

Cheating/plagiarism will not be tolerated on any work. Academic dishonesty on the final exam may result in an **F** for the course. A first occurrence will result in a grade of 0 on the assignment for all concerned parties as well as an Academic Dishonesty form being filed with the Dean's Office. A second occurrence will result in a grade of **F** for the course for the concerned parties and a second Academic Dishonesty form being filed. (See also the University's policy on Academic Honesty at <http://codeofconduct.gsu.edu/>.)

Copyright and Honesty Clause

All content created in this course, including videos, handouts, etc., may be used only by students enrolled in the course for purposes relating to the course. The selling, sharing, publishing, presenting, or distributing of instructor-prepared course lecture notes, videos, audio recordings, or any other instructor-produced materials from any course for any commercial or non-commercial purpose is strictly prohibited, unless explicit written permission is granted in advance by the course instructor. This includes posting any materials on websites such as Chegg, Course Hero, OneClass, Stuvia, StuDocu, and other similar sites. Unauthorized sale or commercial or non-commercial distribution of such material is a violation of the instructor's intellectual property and the privacy rights of students attending the class, and is prohibited. Failure to abide by these limitations constitutes a violation of the Policy on Academic Honesty and will be treated accordingly.

Students Conduct Guidelines

Appropriate conduct is expected from all students. Arrive on time to classroom, and do not leave the classroom early. If you must leave early for some reason, please inform me prior to class. Text messaging, instant messaging, emailing, etc. during class is strictly prohibited and is grounds for dismissal. If you are using your cell phone or computer or another device for tasks that are not math related, or talking, or otherwise disrupting students, you will be asked to leave. After the third incident you will be administratively removed from the class (as per the Student Handbook).

Remark

The course syllabus provides a general plan for the course; deviations may be necessary.